

Product Sentiment Analyzer

Production Deployment & Local Setup Guide

1. System Architecture Overview

The Product Sentiment Analyzer is a full-stack web application consisting of three main components:

- **Frontend Interface:** Built with React (Vite) and Recharts, hosted statically on Vercel or Netlify.
- **Backend API:** Built with Flask (Python), integrating VADER sentiment analysis and the web crawler module, hosted on Render.
- **Database:** MongoDB cluster, self-hosted on AWS EC2 (Ubuntu Linux) to store products, reviews, and sentiment logs.

2. Local Setup Guide

Follow these steps to run the complete system locally:

Step 2.1: Python Backend Setup

Create a virtual environment and install packages:

```
python -m venv .venv
.venv\Scripts\activate (Windows) or source .venv/bin/activate (Mac/Linux)
pip install pandas pymongo vaderSentiment Flask Flask-Cors python-dotenv BeautifulSoup4 tqdm
```

Step 2.2: Seeding Local MongoDB

Verify your local MongoDB server is running on port 27017, and run the seeding script:

```
python database/database.py
```

Step 2.3: Starting Backend and Frontend Servers

Start Flask API:

```
python backend/app.py
```

In another terminal, build and launch the React UI:

```
cd frontend
npm install
npm run dev
```

3. Production Deployment Step-by-Step

Step 3.1: Host MongoDB on AWS EC2

1. Launch an Ubuntu Server EC2 instance on AWS (t2.micro is free-tier eligible).
2. In your EC2 Instance Security Group, add an Inbound Rule for Custom TCP, Port 27017, open to Custom Source (or Anywhere 0.0.0.0/0).
3. SSH into the Ubuntu VM and install MongoDB:
`sudo apt update && sudo apt install -y mongodb`
4. Configure MongoDB to allow remote connections:
Open config: `sudo nano /etc/mongodb.conf`
Change: `bind_ip = 127.0.0.1` to `bind_ip = 0.0.0.0`
5. Restart service: `sudo systemctl restart mongodb`

Step 3.2: Host Backend on Render.com

1. Put your code on GitHub and sign up on Render.com.
2. Create a new Web Service and link your repository.
3. Set Build Command: `pip install -r backend/requirements.txt`
4. Set Start Command: `gunicorn backend.app:app`
5. Under Environment Variables, add:
`MONGO_URI = mongodb://[YOUR-EC2-PUBLIC-IP]:27017/`
`DATABASE_NAME = ProductSentimentDB`
6. Copy the live API URL generated by Render.

Step 3.3: Host Frontend on Vercel or Netlify

1. Update the API Base URL in `frontend/src/context/ProductContext.jsx` to point to your new Render Backend API URL.
2. Commit and push changes to GitHub.
3. Connect Vercel to your GitHub repo, select the root directory as `frontend`, and click Deploy.
4. Vercel will output a live URL (e.g. `https://your-app.vercel.app`).